

Final Exam: Micro 629 - Fall 2009

Texas A&M University

December 14th, 2009

“An Aggie does not lie, cheat, or steal or tolerate those who do.”

EXAM RULES

- Your interpretation of the questions makes part of the exam. Thus, no questions will be answered during the exam.
- No personal items (bags, cellphone, books, notebooks, calculators, etc.) will be allowed with you during the exam.

1. (25/100) Let $\succsim$ be the rational preference on $\mathbb{R}_+^2$ represented by the utility function

$$u(x_1, x_2) \equiv \min\{x_1, x_2\}.$$

- (a) (1/25) Draw the indifference curve that passes through $(1, 2)$.
 - (b) (8/25) Calculate the demand function associated with this preference, i.e., for each $p \equiv (p_1, p_2) \in \mathbb{R}_{++}^2$ and each $w \in \mathbb{R}_+$, calculate $x_1(p, w)$ and $x_2(p, w)$.
 - (c) (4/25) Calculate $S_{12}(p, w)$ and $S_{21}(p, w)$. (4/25) Interpret the result.
 - (d) (3/25) State the uncompensated law of demand. (5/25) Prove that x satisfies the uncompensated law of demand.
2. (25/100) Consider the two expected utility preferences over $(0, +\infty)$ with vNM indices

$$u(x) = \frac{x^\alpha}{\alpha}, \text{ and } v(x) = \frac{x^\beta}{\beta} \text{ where } 1 \geq \alpha \geq \beta > 0.$$

- (a) (10/25) Is one of these agents more risk averse than the other?
- (b) (7.5/25) Define what is the certainty equivalent of a lottery F for an Expected Utility Maximizer.
- (c) (7.5/25) Argue that for each lottery F ,

$$v^{-1} \left(\int v(x) dF(x) \right) \leq u^{-1} \left(\int u(x) dF(x) \right).$$

Interpret this inequality (recall that for each $y \in \mathbb{R}_+$, $u^{-1}(y) = x$ such that $u(x) = y$.)

3. (25/100) Assume that the consumption space of each agent is $\mathbb{R}_+^L$. Let

$$e \equiv ((\succsim_i)_{i \in N}, (Y_j)_{j \in J}, (\omega^i)_{i \in N}, (\theta_j^i)_{i \in N, j \in J}),$$

be a competitive economy.

- (a) (10/25) Define what is a competitive equilibrium for e .
- (b) (15/25) Let $x \equiv (x^i)_{i \in N}$ be a list of consumption bundles, one for each agent. We say that x is envy-free if no agent prefers the consumption of any other agent at x to her own consumption. That is, for any two agents $\{i, k\} \subseteq N$, $x^i \succsim_i x^k$. Assume that all agents

have equal endowments, i.e., for any two agents $\{i, k\} \subseteq N$, $\omega^i = \omega^k$. Assume also that agents have equal shares on each firm, i.e., for each firm $j \in J$ and for any two agents $\{i, k\} \subseteq N$, $\theta_j^i = \theta_j^k$. Prove that if (x, y, p) is a competitive equilibrium for e , then x is envy-free.

4. (25/100) Let $X \equiv \mathbb{R}_+^L$. Let $\succsim$ be a monotone and continuous rational preference relation on $\mathbb{R}_+^L$. Suppose that $\succsim$ is such that there is at least one strict preference statement, formally, that there are $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \succ y$.
 - (a) (20/25) Is there a discontinuous utility representation of $\succsim$? Justify your answer.
 - (b) (5/25) Would your answer change if we do not assume that there is at least one strict preference statement? Justify your answer.
5. (Bonus 25/100) A preference $\succsim$ on $\mathbb{R}_+^L$ is homothetic if for any $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \sim y$ and each $\alpha \in \mathbb{R}_+$, we have that $\alpha x \sim \alpha y$. Assume that $\succsim$ is rational, homothetic, and continuous but not necessarily monotone.

Suppose that $x \in \mathbb{R}_+^L$ is such that $0 \succ x$. Prove, without assuming that the preference can be represented by a homogeneous function, that for each $\alpha \in (0, 1)$, $\alpha x \succ x$.